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Comparing teaching methods on novice nurses' and medical residents' resuscitation teamwork knowledge and team performance: A longitudinal randomized controlled trial

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ABSTRACT

Aim: This study examined the short-term and long-term effects of three resuscitation teaching methods on teamwork knowledge and team performance among novice nurses and medical residents.
Background: Effective resuscitation training is essential for team performance in novice clinical teams.
Design: A single-blind, stratified randomized controlled trial with repeated measures.
Methods: A total of 124 nurses and medical residents within their first 2 years of employment in Taiwan between August and December 2022 were randomly assigned to board-game training, high-fidelity simulation, or interactive lectures (control group). Participants received 3-hour TeamSTEPPS teamwork training with four standardized scenarios. Teamwork knowledge and team performance were assessed at pretest, posttest and 3-month follow-up. Data were analyzed using generalized estimating equations (GEE). The study was registered at ClinicalTrials.gov (NCT05302414).
Results: Ninety-six participants (24 teams) completed the study. Posttest teamwork knowledge was higher in the control group than in the simulation group ($p = 0.04$). From pretest to posttest, the control group showed significant knowledge gains ($p = 0.003$), while the simulation group improved in team performance ($p = 0.012$). GEE analysis indicated a significantly lower increase in knowledge scores for the simulation group compared with the control group ($B = -1.09$, $p = 0.04$). Board-game group feedback emphasized fun, relaxation and communication.
Conclusions: Simulation training was most effective in improving team performance, interactive lectures enhanced teamwork knowledge and board-game training provided a positive learning environment that may

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increase learner engagement. Educators should select the most effective method based on the specific learning objectives developed.

1. Introduction

In healthcare settings, providers must closely collaborate to deliver care, especially in critical and emergent situations (Sangaleti et al., 2017). Research has consistently demonstrated that effective interprofessional collaboration significantly enhances clinical outcomes and improves patient safety (McCutcheon et al., 2020; Webster et al., 2024). Teamwork is a core and essential component of interprofessional collaboration, characterized by shared goals, role clarity, interdependence and collective decision-making (Canadian Interprofessional Health Collaborative, 2024; Reeves et al., 2010). Resuscitation demands rapid coordinated actions in high-stakes crises like cardiac arrest, where no single profession can handle all tasks. Both the American Heart Association and European Resuscitation Council integrate teamwork and interprofessional collaboration into resuscitation training to ensure tasks are divided efficiently among diverse experts and thus improve survival rates (Cheng et al., 2020; Soar et al., 2021). Poor teamwork, often due to unclear leadership, ambiguous task assignments, or communication failures among inexperienced or novice medical residents and nurses, frequently jeopardizes patient safety and poses significant barriers to effective healthcare delivery (Nallamothu et al., 2018; O'Connor et al., 2016; Umberfield et al., 2019). For teams to perform effectively, teams must possess substantial teamwork knowledge and demonstrate appropriate teamwork performances to foster effective collaboration, enhance patient safety and optimize care.

To meet these educational needs, various teaching methods have been employed to develop effective teamwork behaviors among healthcare professionals. In the past, interactive lectures were designed to build conceptual understanding of roles, leadership and communication (Cooke, 2016). However, systematic reviews have increasingly supported simulation-based training as an effective method for enhancing interprofessional team performances in resus (Abildgren et al., 2022; Buljac-Samardzic et al., 2020; Sezgin and Bektas, 2023). Simulation-based learning, particularly high-fidelity simulation, allows learners to apply teamwork principles in realistic, high-pressure settings, enhancing coordination and decision-making (Abildgren et al., 2022; Cheng et al., 2020). However, despite its benefits, high-fidelity simulation as high costs, is time-consuming (both in terms of educator training and simulation scenario design) and has high technical support needs (Adamson, 2010; Al-Ghareeb and Cooper, 2016; Krishnan et al., 2017), which limit its use.

It is important for the nursing and medical professions to collaborate through an effective, convenient and cost-effective training method for team resuscitation. Gamified and board game-based teaching might be a potential solution that has been used in emergency resuscitation situations to enhance knowledge acquisition, improve interpersonal interactions and motivate improvement (Cheng et al., 2020; Cutumisu et al., 2019; Noda et al., 2019). A well-designed game-based learning tool can foster students' cognitive, behavioral, affective and sociocultural engagement during teamwork training (Friedrich et al., 2019; Plass et al., 2015). Overall, three instructional approaches, including card-based board game learning, high-fidelity simulation-based learning using computerized patient manikins and interactive lecture-based learning delivered through structured discussions, have all been used in resuscitation training. The board game and high-fidelity simulation-based learning both provide experiential and interactive learning that fosters teamwork competence. However, few studies have simultaneously compared instructional approaches and examined the retention of teamwork knowledge and performance, which limits educators in deciding on the most appropriate teaching method to use.

Therefore, the purposes of this study were to examine and compare the short-term effects and 3-month retention of novice nurses' and medical residents' (1) teamwork knowledge; and (2) team performance competencies (e.g., leadership, communication, mutual support and situation monitoring) in resuscitation scenarios trained in board game-based learning, high-fidelity simulation-based learning and interactive lecture-based learning methods. Accordingly, this study addressed the following research questions: (1) What are the different effects of resuscitation teamwork, knowledge and team performance in board game-based learning, high-fidelity simulation-based learning (using computerized simulators) and interactive lecture-based learning? and (2) What are the 3-month retention levels of these three teaching methods on resuscitation teamwork knowledge and performance?

2. Methods

2.1. Study design and participants

A prospective, single-blind, three-arm randomized controlled trial with a repeated-measures design was conducted at an 800-bed teaching hospital in northern Taiwan between August and December 2022. Medical residents and nurses were recruited through convenience sampling. Novice nurses and medical residents were targeted because they participate in resuscitation teams but lack confidence and experience in teamwork under high-stress conditions (Sukor and Sabardin, 2025; Wang et al., 2024). Participants were eligible if they were aged 20 years or older and were postgraduate year medical residents or nurses in their first 2 years of employment. Individuals were excluded if they did not provide direct patient care or worked in pediatric, obstetric, psychiatric, palliative care, or long-term care wards. Prior to the intervention, participants received standardized electronic pretraining materials emailed by administrative staff one week in advance. Based on the 2020 American Heart Association Guidelines for Cardiopulmonary Resuscitation and Emergency Cardiovascular Care and the TeamSTEPPS 2.0 curriculum, the materials ensured consistent baseline knowledge and covered resuscitation procedures including initial assessments, cardiac arrest and airway management, cardioversion, return of spontaneous circulation and four teamwork competencies: leadership, communication, mutual support and situation monitoring. The study outcomes included teamwork knowledge, team performance assessed through a paper-based test, high-fidelity simulation and qualitative feedback. These outcomes were measured at three time points: pretest, posttest and 3-month follow-up.

2.2. Randomization

Stratified random sampling was used. Eligible participants were first divided into two strata based on their professional role (medical residents and nurses). The researcher configured an Excel file to automatically generate random codes for group assignments, with separate settings for each stratum. Administrative staff entered the names of medical residents and nurses and the program assigned them to groups using preset functions. This ensured a balanced number of interprofessional members on each team and equal numbers of teams across the teaching method groups. Each learning team consisted of one medical resident and three nurses. The designated teaching method for each team was placed in a sealed envelope by the administrative staff and revealed to participants at the training site on the day of the activity.

2.3. Sample size

The sample size for this study was determined using Power Analysis

and Sample Size Software, vers. 23.0.2(NCSS, LLC., 2023). Calculations were based on generalized estimating equations tests for the treatment and disease interaction of multiple groups in a repeated-measures design. Effect sizes were derived from group means of 19.88, 19.92 and 19.50 and corresponding standard deviations (SDs) of 0.19, 0.14 and 0.52, as referenced from Hobgood et al. (2010). With a significance level of 0.05 and a desired statistical power of 0.80, the required sample size was estimated to be 57 participants. Considering a potential 20% missing data rate, at least 72 participants were needed. A total of 124 participants were ultimately included in the study.

2.4. Theoretical background

This intervention study was grounded in Bandura's social learning theory, which emphasizes learning through observing and modeling behaviors, attitudes and emotional responses of others (Bandura, 1977) and Kolb's experiential learning theory, which views learning as the transformation of experience (Kolb, 1984). These theoretical frameworks were applied to the design of the two interprofessional intervention programs, board game-based learning and high-fidelity simulation-based learning, both of which integrated social interactions and experiential learning. Medical residents and nurses collaboratively worked through critical patient scenarios, observed each other's roles, actions and consequences and engaged in various team-based activities to gain practical experience. Specifically, participants engaged in resuscitation scenarios designed for experimental groups as concrete experiences, reflected on teamwork challenges during debriefing sessions (reflective observations), identified key teamwork principles related to the TeamSTEPPS framework (abstract conceptualization) and applied improved teamwork and communication strategies in subsequent scenarios (active experimentation). Post-activity debriefings followed the gather-analyze-summarize (GAS) model (Phrampus, O'Donnell, 2013) to structure participants' reflections on teamwork challenges and identify areas for future improvement. In the gathering phase, participants shared their thoughts and described the scenario. During the analyzing phase, facilitators encouraged reflection on teamwork behaviors (e.g., "I noticed that during…, what approach did you take?"). In the summarizing phase, participants discussed what the team did well, what should be maintained and what could be improved in future situations. The control group, which received interactive lecture-based learning, participated in discussions and received instructor feedback.

2.5. Resuscitation teamwork learning content

The study was conducted in 2022 and the TeamSTEPPS 2.0 curriculum (Agency for Healthcare Research and Quality [AHRQ], 2019) was used to develop the training content for all three training groups. The AHRQ permits the free, unrestricted use of TeamSTEPPS 2.0 materials. The 15 practical skills emphasize leadership, communication, mutual support and situation monitoring and these were all incorporated into the scenarios. To address specific needs of resuscitation environments, additional content was integrated into the training, requiring participants to perform essential tasks such as setting up an emergency environment, prioritizing multiple responsibilities and supporting team members. Four standardized clinical case scenarios, including hypoxia, hyperkalemia, hypovolemia and acute myocardial infarction, were developed and validated by three clinical experts with over 10 years of emergency care experience and were applied across all three training groups:

• Scenario 1 (hypoxia): A 70-year-old male who was hospitalized for pneumonia presents with rapid breathing and dyspnea. • Scenario 2 (hyperkalemia): A 55-year-old female in the emergency department is diagnosed with acute kidney injury and presents with no pulse. • Scenario 3 (hypovolemia): An 80-year-old male who was hospitalized for suspected gastrointestinal bleeding presents with confusion. • Scenario 4 (acute myocardial infarction): A 65-year-old female in the emergency department is diagnosed with acute coronary syndrome and presents with no pulse. These four scenarios were used as learning cases in resuscitation teamwork training, allowing participants to assess situations, communicate and coordinate interventions, with outcomes provided based on their actions to guide real-time adjustments. The scenarios were designed to target key teamwork skills such as task prioritization, delegation and effective communication.

2.6. Intervention groups and control group

This study involved three training groups (see Supplementary File l): board game-based learning (experimental group I), high-fidelity simulation-based learning (experimental group II) and interactive lecture-based learning (control group). To control variations in teaching content, all groups used the same teaching materials (TeamSTEPPS 2.0), case scenarios and a standardized teaching manual. Each group was instructed by two well-trained instructors with at least 10 years of emergency clinical experience and 8 years of teaching experience. To ensure instructional consistency, all instructors followed a standardized manual when implementing the teaching activities. The standardized teaching manual provided a structured overview of resuscitation teamwork training, including the course topic, purpose, training duration, teaching methods, instructional and learning objectives, four standardized clinical case scenarios, session procedures and post-session activities, such as debriefing, feedback and discussion. The learning content was based on the TeamSTEPPS 2.0 framework and the manual was distributed to facilitators of different groups to ensure instructional consistency. Each training group's 3-hour session aligned with the TeamSTEPPS course duration (Welsch et al., 2018) and institutional scheduling feasibility. All groups completed four scenarios at the Medical Simulation Education Center, where the intervention and control courses were conducted concurrently over 3 consecutive days.

2.6.1. Board game-based learning

An educational board game, *Code Team* (see Supplementary File 2), was developed and validated by three clinical experts with over 10 years of emergency care experience. Participants first watched a standard tutorial video to learn the rules and procedures. Each team then played four 12-minute rounds based on standardized clinical scenarios. The game used Patient Information, Task and Management cards to structure case assessment, clinical decision-making and the application of relevant TeamSTEPPS competencies. The game included ten emergency tasks representing key clinical steps and participants collaborated to complete all tasks within the time limit through mechanics designed to apply teamwork competencies. After each round, a facilitator conducted a structured 25-minute debriefing using the GAS model and a learning sheet to review key events and discuss strategies that enhanced teamwork and effective patient management.

2.6.2. High-fidelity simulation-based learning

The high-fidelity simulation was conducted in a simulated emergency care room using computerized human patient manikins capable of real-time physiological responses. The instructor controlled the computerized simulator's physiological responses based on standardized clinical case scenarios, allowing the learning team to assess and respond to real-time changes while practicing teamwork behaviors. The setup replicated a clinical environment, placing the manikin on a hospital bed and equipping the space with medical devices such as a vital sign monitor, a defibrillator and other medical supplies. Each learning team was oriented to the simulation environment and then participated in four scenarios/rounds. Each round/scenario using a computerized simulator was 12 min long, followed by a 25-minute debriefing session conducted by a facilitator, allowing students to reflect on their performance and identify areas for improvement. All debriefing sessions followed the GAS model, focusing on teamwork reflection and strategy

improvement.

2.6.3. Interactive lecture-based learning

An interactive teaching method which has been used in resuscitation topics at the study site served as the control group in this study. Instructors used structured presentation slides to facilitate interactive discussions. The first 20 min were allocated to ice-breaking activities, followed by four 35-minute sessions, each focusing on a specific teamwork concept. Each session focused on one of the four teamwork competencies paired with a corresponding standardized clinical case scenario. Instructors facilitated interactive discussions, role-playing and feedback to help participants apply the TeamSTEPPS principles during resuscitation practice. The course concluded with a 20-minute comprehensive exercise integrating all teamwork competencies, followed by instructor-led feedback.

2.7. Rater training

Two raters, including a physician with 5 years of emergency clinical experience and a nurse practitioner with 10 years of emergency clinical experience, were invited to assess each team's performance at the pretest, posttest and 3-month follow-up. Both of them had taught resuscitation based on the TeamSTEPPS principles for more than 5 years. Additionally, they underwent rater training based on the method proposed by Eppich et al. (2015). First, the research team selected three recordings from past resuscitation simulation trainings, which represented poor, mixed and excellent team performances. Second, both raters independently watched these three recordings and assessed team performances using the Team Performance Observation Tool (TPOT) developed under the TeamSTEPPS program by the Agency for Healthcare Research and Quality (AHRQ) and the Department of Defense (DoD). Third, the two raters independently scored and then discussed any discrepancies until they reached satisfactory agreement (intercorrelation coefficient (ICC) = 0.896), before they could grade the team performances for this study.

2.8. Outcome measurements

Demographic data were collected during the pretest. Teamwork knowledge and team performance were measured at three time points: pretest, posttest and 3-month follow-up. Qualitative feedback was also obtained at posttest to better understand participants' learning experiences and suggestions for course improvement.

2.8.1. Demographic data

The collected demographic data included self-reported gender (male/female), age, educational level, work experience, work site, resuscitation experience and ACLS/BLS certification status.

2.8.2. Teamwork knowledge

The Knowledge of Teamwork questionnaire was developed based on TeamSTEPPS Learning Benchmarks (Agency for Healthcare Research and Quality AHRQ, 2019) which has 23 multiple-choice items, each with five answer choices and only one correct answer. Participants received one point for each correct response, resulting in a total score ranging from 0 to 23, with higher scores indicating a more-comprehensive understanding of teamwork. The test underwent back translation and a content validity assessment. The item-level content validity index (I-CVI) was 0.8 and the scale-level content validity index (S-CVI/AVE) was 0.99, indicating high appropriateness and clarity of item contents. The reliability coefficient, as assessed by the Kuder-Richardson-20 test, was 0.81.

2.8.3. Team performances

Each learning team managed a patient resuscitation in a standardized computerized simulator high-fidelity scenario at the pretest, posttest and 3-month follow-up and all teams' performances were video-recorded for assessment. A team's performance was assessed using the TPOT developed under the TeamSTEPPS program by the AHRQ and DoD. TPOT is a 23-item rating checklist covering five categories: team structure (four items), leadership (six items), communication (four items), situation monitoring (five items) and mutual support (four items). Each item is scored from 1 (very poor) to 5 (excellent), resulting in a cumulative score ranging from 23 to 115, with higher scores indicating better team performance. Inter-rater reliability scores of the scale were 0.81, 0.80 and 0.82 for the pretest, posttest and 3-month follow-up, respectively, which were considered acceptable (Graham et al., 2014). Two trained raters, blinded to the teams' assigned teaching methods (single-blind), independently evaluated team performances by reviewing the video recordings.

2.8.4. Qualitative feedback

Participants' learning experiences and overall perspectives on the training course were collected at the post-test. Two open-ended questions, 1) a description of their experiences in the assigned resuscitation team collaboration training group and 2) suggestions for improvement, were asked.

2.9. Data analysis

Categorical variables in the study are presented as frequencies and percentages, while means and SDs were used to summarize continuous variables. Baseline demographic characteristics were analyzed using the Chi-squared test. Since the data were not normally distributed, nonparametric analyses were employed, with median and interquartile range values calculated (Habibzadeh, 2024). Group differences were tested using the Kruskal-Wallis H test. Within-group changes across the three time points were assessed with the Friedman test, followed by Bonferroni-adjusted Wilcoxon post-hoc tests ($p < 0.016$) (Armstrong, 2014). Generalized estimating equations (GEEs) were used to examine trends across the three time points and explore group differences. Additionally, the GEE is an appropriate method for managing correlated repeated-measures data, estimating population-averaged effects and handling unbalanced or missing observations without requiring strict distributional assumptions (Ballinger, 2004; Liang and Zeger, 1986). Effect sizes were reported as R^2 in the results table. All statistical analyses were performed using IBM SPSS vers. 20 (IBM, Armonk, NY, USA), with statistical significance determined by two-tailed tests at $p < 0.05$. A word cloud module in Python 3.10 was used to analyze participants' textual feedback to identify high-frequency words. The raw data were preprocessed to enhance accuracy and reduce errors (Vijayarani et al., 2015). In addition, this study established a key term frequency table based on the Interprofessional Education Collaborative (IPEC) Core Competencies Framework, categorizing participants' feedback into four domains: Values and Ethics, Roles and Responsibilities, Communication and Teams and Teamwork to complement the qualitative analysis.

2.10. Ethical considerations

This study was approved by the institutional review board (N201912063) and registered at ClinicalTrials.gov (ID: NCT05302414). The teaching material used in this study, TeamSTEPPS 2.0, is a publicly available curriculum released by AHRQ (2019). The research team introduced the study to participants and answered research-related questions before obtaining informed consent. All participants enrolled voluntarily and were informed that they could withdraw from the study and data collection at any time.

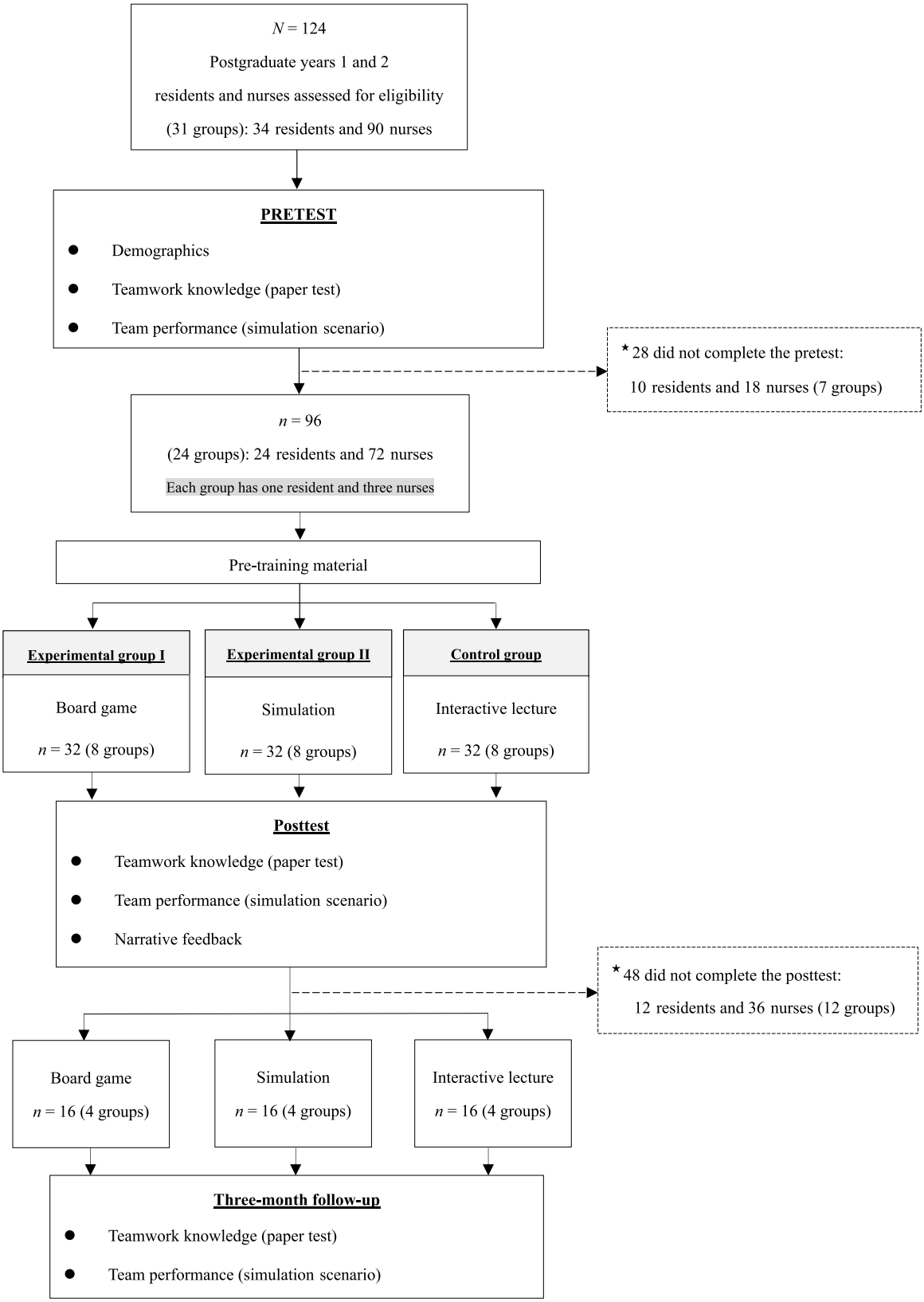

Fig. 1. Study design through each stage. Notes: *n* represents the number of participants (with the number of teams indicated in parentheses). Sessions were held over four days for the pre-, post-, and three-month follow-up tests, and three consecutive days for the intervention courses. The approximate time intervals: 13–23 days between the pre-test and intervention, one week between the distribution of pre-training materials and the intervention, and about 90 days for follow-up. ★ The reasons for non-participation included staff shortages and scheduling conflicts, with nine participants unable to attend the pretest due to being diagnosed with COVID-19.

Table 1
Demographics of participating subjects ($N = 96$, 24 residents, 72 nurses).

Variable	Board game ($N = 32$)	Simulation ($N = 32$)	Interactive lecture ($N = 32$)	*n*	*p*
	n (%)	*n* (%)	*n* (%)		
Age (years), mean ± SD	25.63 ± 4.16	24.31 ± 2.49	24.97 ± 3.04		0.38[a]
Work experience, (months)					0.09
0–6	6 (18.8%)	15 (46.9%)	13 (40.6%)	34	
7–12	11 (34.4%)	13 (40.6%)	11 (34.4%)	35	
13–18	7 (21.9%)	2 (6.2%)	4 (12.5%)	13	
19–24	8 (25.0%)	2 (6.2%)	4 (12.5%)	14	
Gender					0.63
Male	6 (18.8%)	5 (15.6%)	8 (25.0%)	19	
Female	26 (81.2%)	27 (84.4%)	24 (75.0%)	77	
Educational level					1.00
Bachelor's or above	30 (93.8%)	31 (96.9%)	31 (96.9%)	92	
Associate or below	2 (6.2%)	1 (3.1%)	1 (3.1%)	4	
Working sites					0.07
Medical ward	4 (12.5%)	8 (25.0%)	6 (18.8%)	18	
Surgical ward	9 (28.1%)	2 (6.3%)	7 (21.9%)	18	
Special unit[Ψ]	8 (25.0%)	8 (25.0%)	2 (6.2%)	18	
Operating room	3 (9.4%)	9 (28.1%)	10 (31.2%)	22	
Others[§]	8 (25.0%)	5 (15.6%)	7 (21.9%)	20	
Resuscitation experience (times after work)					0.31
1	6 (18.8%)	5 (15.6%)	2 (6.2%)	13	
≤ 5	9 (28.1%)	8 (25.0%)	7 (21.9%)	24	
≥ 6	4 (12.5%)	1 (3.1%)	1 (3.1%)	6	
None	13 (40.6%)	18 (56.2%)	22 (68.8%)	53	
ACLS/BLS certification					0.13
Yes	24 (75.0%)	30 (93.8%)	28 (87.5%)	82	
No	8 (25.0%)	2 (6.2%)	4 (12.5%)	14	

Notes: Chi-squared test; [a] = Kruskal-Wallis H test; * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$, two-tailed; SD, standard deviation; ACLS, Advanced Cardiovascular Life Support; BLS, Basic Life Support; [Ψ]= Intensive Care Unit and Emergency Room; [§] = Outpatient, Pathology, Rehabilitation, etc.

3. Results

3.1. Participant characteristics across the three groups

The study initially enrolled 124 participants. Of these, 96 successfully completed the intervention course, along with the pretest and posttest. Each intervention group was composed of 32 participants, which included eight medical residents and 24 nurses. Due to scheduling conflicts during the outbreak of COVID-19, only 48 participants completed the 3-month follow-up, which consisted of four teams (four medical residents and 12 nurses) in each group (Fig. 1).

A comparative analysis among the three groups showed no statistically significant differences between groups. Detailed demographic profiles of the participants are presented in Table 1.

3.2. Differences in teamwork knowledge and team performance between the groups

Table 2 presents differences in teamwork knowledge and performance among the three groups at the pretest, posttest and 3-month follow-up. No statistically significant differences were found in teamwork knowledge or performance at the pretest, nor in team performance at the posttest and 3-month follow-up. However, at the posttest, the interactive-lecture group had a significantly higher knowledge score (median [first quartile (Q1), Q3]: 20.00 [17.00, 21.00]) than both the simulation group (18.00 [16.00, 19.00]) and board-game group (19.00 [17.00, 20.00]). Furthermore, post-hoc pairwise comparisons using Dunn's test indicated that the knowledge score was significantly higher in the interactive-lecture group compared with the high-fidelity simulation group (median: 20.00 vs. 18.00, $p = 0.04$).

3.3. Changes in teamwork knowledge and team performance within each group

Table 3 demonstrates changes of teamwork knowledge and team performance over time. From the pretest to posttest, teamwork knowledge significantly improved in the interactive lecture group, with median (Q1, Q3) scores increasing by 1.00 (0.00, 3.00; p = 0.003). From the pretest to posttest, team performance significantly improved in the simulation group, with median (Q1, Q3) scores increasing by 11.75 (10.00, 16.38; p = 0.01). However, no statistically significant changes were observed in teamwork knowledge or team performance in the board-game group.

3.4. Effectiveness of retention outcomes on teamwork knowledge and team performances of the intervention among groups

Table 4 presents the 3-month follow-up outcomes of teamwork knowledge and performance across three different groups. In terms of teamwork knowledge, the GEE analysis revealed that participants showed improvement after the training courses (B = 1.34, 95% CI: 0.56–2.12) and these gains were maintained even 3 months later (B = 1.26, 95% CI: 0.13–2.39). Despite the overall 3-month retention of the resuscitation teamwork knowledge, the finding suggests that at the posttest, the simulation group's teamwork knowledge score was lower than that of the interactive lecture group (B = −1.09, 95% CI: −2.18 to −0.01, $p = 0.04$). The coefficient of determination (R^2) of the analysis on team knowledge was 4.34%. However, the simulation group had improved its team performance by the posttest (B = 11.81, 95% CI: 5.75–17.87) and had even maintained it at the 3-month follow-up (B = 9.72, 95% CI: 5.25–14.18) compared with the interactive lecture group. The coefficient of determination (R^2) of the analysis on team behavior was 33.1%. Considering potential attrition bias, multiple imputation was performed. The imputed GEE results were consistent with the original findings, showing significant time effects with improvements in teamwork knowledge at the posttest and 3-month follow-up. No significant group or interaction effects were found, indicating comparable learning outcomes across the three instructional approaches (See Supplementary File 3).

3.5. Participants' qualitative feedback word cloud results

The word cloud created from participants' course feedback reflects their experiences (Supplementary File 4). Each learning group reported both positive and negative experiences. The board game group generated 116 words, with "fun" ($n = 6$), "relaxed" ($n = 5$) and

Table 2
Teamwork knowledge and team performance outcomes of the pretest, posttest, and 3-month follow-up between groups.

Outcome / Groups		Pretest			Posttest			Three-month follow-up	
		Median (Q1, Q3)	*p*		Median (Q1, Q3)	*p*		Median (Q1, Q3)	*p*
Teamwork knowledge	***n***		0.99	***n***		0.04* #	***n***		0.96
Board game	32	18.00 (16.00, 19.00)		32	19.00 (17.00, 20.00)		16	19.00 (17.25, 19.75)	
Simulation	32	17.00 (16.00, 19.00)		32	18.00 (16.00, 19.00)		16	19.00 (18.00, 19.75)	
Interactive lecture	32	17.00 (15.25, 19.75)		30	20.00 (17.00, 21.00)		16	19.50 (16.00, 20.00)	
Team performance	**Team**		0.11	**Team**		0.57	**Team**		0.69
Board game	8	53.00 (47.00, 59.00)		8	57.75 (53.38, 62.13)		4	62.50 (46.63, 67.50)	
Simulation	8	39.00 (37.63, 48.50)		8	56.50 (49.63, 59.88)		4	56.75 (44.38, 62.75)	
Interactive lecture	8	39.25 (35.75, 53.50)		8	51.50 (46.88, 65.00)		4	57.25 (49.13, 68.00)	

Notes: In the team performance, each time had one resident and three nurses. Kruskal-Wallis H test, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$, two-tailed. # Dunn's test revealed
a test statistic of 17.132 ($p = 0.04$) between the simulation and interactive lecture groups. Q1, first quartile; Q3, third quartile.

Table 3
Teamwork knowledge and team performance outcomes of the pretest, posttest, and 3-month follow-up within each group.

Outcome / Group		Posttest from Pretest			Three-month follow-up from Posttest			Three-month follow-up from Pretest	
		Median (Q1, Q3)	*p*		Median (Q1, Q3)	*p*		Median (Q1, Q3)	*p*
Teamwork knowledge	***n***			***n***			***n***		
Board game	32	0 (-1.00, 2.00)	0.31	16	0 (-1.00, 1.00)	0.75	16	0.50 (-1.00, 2.75)	0.22
Simulation	32	0 (-1.00, 1.00)	0.44	16	0.50 (-0.75, 1.75)	0.26	16	1.00 (0.00, 2.00)	0.02
Interactive lecture	30	1.00 (0.00, 3.00)	0.003*	16	0.00 (-2.00, 1.00)	0.57	16	1.00 (0.00, 3.50)	0.07
Team performance	**Team**			**Team**			**Team**		
Board game	8	4.75 (0.88, 12.38)	0.03	4	0 (-4.25, 6.13)	0.85	4	9.50 (0.75, 13.38)	0.14
Simulation	8	11.75 (10.00, 16.38)	0.012*	4	1.75 (-5.25, 3.13)	0.72	4	13.25 (6.63, 15.75)	0.07
Interactive lecture	8	13.00 (6.88, 15.00)	0.03	4	-4.75 (-16.50, 11.88)	0.47	4	8.25 (7.50, 10.88)	0.07

Notes: In the team performance, each time had one resident and three nurses. Wilcoxon signed-rank test (Bonferroni-adjusted), two-tailed, $p < 0.016$. Q1, first quartile; Q3, third quartile.

"communication" ($n = 4$) being the most frequently mentioned positive feelings. Negative feelings included "complex rules" and "time-consuming." The simulation group produced 179 words, featuring "actual" ($n = 9$), "practice" ($n = 8$) and "exercise" ($n = 8$) as the most common favorable terms. Negative feelings included "unfamiliarity," "stress," and "distraction." The interactive lecture group generated 137 words, with "scenario" ($n = 10$), "discussion" ($n = 7$) and "exercise" ($n = 6$) leading. Negative feelings included "lack of immersion" and being "purely discussion-based." A frequency table of key terms was developed to complement the word cloud based on the IPEC Core Competencies Framework (see Supplementary Files 5). Some feedback could not be categorized under the IPEC domains, as it reflected participants' learning experiences or emotions, such as "board game learning was fun," "the simulation felt like an actual and stressful experience," and "the lecture lacked immersion." The word cloud illustrated the learning atmosphere and engagement, while the frequency table provided a structured categorization of key terms, together offering a comprehensive understanding of the qualitative findings.

4. Discussion

4.1. Summary of key findings

The study findings showed that different teaching methods had unique strengths and limitations and thus each was better suited for specific situations and goals. Interactive lectures are most effective in improving team knowledge, while simulation is most effective in enhancing team performance. There were no statistical differences in teamwork knowledge or performance when using board game-based learning compared with the interactive lectures and simulations. Although board game–based learning did not yield statistically significant improvements, its capacity to enhance engagement and reduce learner stress suggests pedagogical value. Accordingly, it may serve as a complementary tool in resuscitation education, particularly in contexts with limited access to simulation resources or when supporting the development of teamwork knowledge and performance among novice nurses and medical residents.

4.2. Comparison and interpretation of teaching methods

Our findings showed that interactive lecture–based learning was particularly effective in improving teamwork knowledge. Previous studies (Ahn and Lee, 2021; Cooke, 2016) also showed that interactive lectures incorporating scenario- or video-based discussions can significantly improve teamwork knowledge.

Since novice healthcare providers often have few opportunities to participate in and perform resuscitation, providing situations where participants have to think about and speak aloud their thoughts really enhances their ability to review the knowledge they have learned and how to apply it in the scenarios provided (Aflah and Rahmani, 2022; Snell, 2009). However, interactive lectures provide limited opportunities for hands-on teamwork practice, which may limit the development of more complex performance skills.

Consistent with previous research (Abildgren et al., 2022; Sezgin and Bektas, 2023), our study found that simulation is an effective teaching method for developing healthcare professionals' teamwork behaviors in resuscitation scenarios. Simulation training provides an opportunity to develop their skills and knowledge through practice and debriefing in safe and realistic scenarios (Buljac-Samardzic et al., 2020; Chamberland et al., 2018). However, despite its effectiveness, simulation is costly and time-consuming, requiring trained instructors, resource allocation and technical and administrative support. These factors limit its application in medical and nursing education (Al-Ghareeb and Cooper, 2016; Krishnan et al., 2017). Given these practical constraints, our study findings suggest that a resuscitation board game could serve as a complementary approach for creating safe and meaningful clinical emergency scenarios that allow learners to form a team, assess, respond to and act in various situations, particularly in settings where simulation

Table 4
Longitudinal outcomes of teamwork knowledge and team performance between different groups.

Outcome	B	SE	95% Confidence interval		p
			Lower	Upper	
Teamwork knowledge (*N*=96)					
Intercept	17.31	0.47	16.39	18.24	< 0.001***
Group[a]					
Board game	< 0.001	0.68	-1.34	1.34	1.00
Simulation	0.19	0.59	-0.98	1.35	0.75
Time[b]					
Posttest	1.34	0.40	0.56	2.12	0.001**
Three-month follow-up	1.26	0.58	0.13	2.39	0.03*
Group × Time[c]					
Board game × Posttest	-0.72	0.69	-2.08	0.64	0.30
Board game × Three-month follow-up	-0.16	0.86	-1.84	1.52	0.86
Simulation × Posttest	-1.09	0.55	-2.18	-0.01	0.04*
Simulation × Three-month follow-up	-0.41	0.67	-1.72	0.91	0.54
Team performance (*N*=24)					
Intercept	42.38	3.27	35.97	48.78	< 0.001***
Group[a]					
Board game	8.69	4.53	-0.19	17.56	0.06
Simulation	-0.25	3.93	-7.95	7.45	0.95
Time[b]					
Posttest	11.81	3.09	5.75	17.87	< 0.001***
Three-month follow-up	9.72	2.28	5.25	14.18	< 0.001***
Group × Time[c]					
Board game × Posttest	-5.75	3.71	-13.03	1.53	0.12
Board game × Three-month follow-up	-2.36	3.41	-9.05	4.33	0.49
Simulation × Posttest	1.75	4.06	-6.21	9.71	0.67
Simulation × Three-month follow-up	2.92	3.49	-3.92	9.76	0.40

Notes: Generalized estimating equations analysis with an exchangeable working correlation matrix.
Effect sizes (R^2): 0.043 for teamwork knowledge; 0.331 for team performance.
SE, standard error. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$, two-tailed.
[a] Reference group: Interactive lecture (Control group).
[b] Reference group: Time (Pretest).
[c] Reference group: Group (Control group) × Time (Pretest).

resources are limited.

Our study results support the notion that students developed team communication skills during a simulation and felt entertained and relaxed while playing our resuscitation board game, which is consistent with previous study findings (Cutumisu et al., 2019; Wang et al., 2024). Although board game–based learning did not produce statistically significant improvements in teamwork knowledge or performance in the present study, previous research suggests that board game-based learning can immerse learners in valuable experiences shaped by the game's characteristics, player behaviors and thematic or contextual elements (Farkas et al., 2020). Compared with simulations, board game-based learning reduces the pressure and anxiety that beginners might face in high-stress resuscitation or emergency scenarios (Cantrell et al., 2017; Judd et al., 2019). The absence of statistically significant improvements in this study may be related to several considerations when implementing board game-based learning. These may include game time management (Boghian et al., 2019), game fluency (Admiraal et al., 2011), players' gaming experience (Ghoman et al., 2020) and potential discrepancies between the game and actual clinical procedures (Kuruca Ozdemir and Dinc, 2022). Board games combine learning content with gameplay elements and the accompanying instructions (e.g., explanations of gameplay and rules) may increase learners' extraneous cognitive load by adding additional information. Effectively addressing these issues can reduce such loads and enhance the overall learning experience (Sewell et al., 2019).

Overall, the three teaching methods exhibited distinct characteristics suitable for various educational contexts. Simulations, although resource-intensive, supported the development of behavioral competencies. Interactive lectures facilitated foundational knowledge acquisition, while board game-based learning provided a pedagogical innovation, fostering a fun, relaxed and communicative learning environment in resuscitation training. Each method demonstrated unique strengths and was deliberately designed to align with learners' needs and instructional goals, thereby optimizing learning outcomes.

4.3. Retention and educational implications

This study found that the training effects were retained at the 3-month follow-up, which aligned with previous findings on the sustained impact of interactive lectures and simulation training (Lisbon et al., 2016; Walker et al., 2014). Knowledge and skills acquired through training tend to decline over time, especially in resuscitation scenarios where skills deteriorate faster than knowledge (Yang et al., 2012). Multiple or booster sessions for cardiopulmonary resuscitation are recommended (Cheng et al., 2020). Evidence also shows that repeated low-dose, high-frequency training supports retention (Monsieurs et al., 2015; Murre and Dros, 2015). High cognitive loads and stressful learning environments can hinder performance (Vogel and Schwabe, 2016). Board game-based learning allows learners to repeatedly engage with teamwork concepts and may facilitate knowledge retention. While resuscitation events can be highly stressful in real clinical situations, repetitive practice in a low-stress environment may help teams develop more-appropriate responses when facing actual emergencies. While recognizing the strengths and limitations of various educational methods and their impacts on working memory, healthcare educators are encouraged to select strategies with care. Integrating diverse learning activities and incorporating booster courses may help support and sustain long-term learning outcomes.

5. Limitations

This study has several limitations that may affect its generalizability. It was conducted in a single hospital in Taiwan and included only two professional roles, which may have limited the applicability of the findings to other contexts. Future evaluations of teaching methods should involve more-diverse settings and professional roles. Convenience sampling may have introduced selection bias and reduced representativeness. In addition, given the data distribution, non-parametric statistical methods were employed. Although these methods generally have lower statistical power, they were appropriate for the nature of the data and provided accurate results. The study was conducted during the COVID-19 pandemic, which led to incomplete follow-up. An attrition analysis using a pooled multivariable regression after multiple imputations revealed no significant demographic effects, with resuscitation experience showing marginal significance ($p = 0.096$). The post-test score significantly predicted teamwork knowledge at the 3-month follow-up. Although attrition did not cause systematic bias, the reduced sample size may have limited the statistical power; therefore, the findings should be interpreted with caut

6. Conclusions

Different teaching methods support different learning outcomes. High-fidelity simulation improves team performance, interactive lectures build teamwork knowledge and board game-based learning enhances engagement and reflection. Integrating multiple teaching methods into resuscitation training may provide a more comprehensive learning experience and contribute to teamwork development and retention of knowledge and skills, particularly for novice nurses and medical residents. Learning objectives should be considered when selecting the most appropriate pedagogy to link knowledge and

behavioral practice. Additionally, routine and booster training sessions are essential for sustaining long-term knowledge and skills in resuscitation.

CRediT authorship contribution statement

Hui-Wen Chen: Writing – review & editing, Writing – original draft, Validation, Methodology, Formal analysis, Data curation, Conceptualization. **Chien-Yu Chen:** Resources, Methodology, Conceptualization. **Guo Shu Liu:** Resources, Methodology, Data curation, Conceptualization. **Pei-Ching Liu:** Writing – review & editing, Methodology, Conceptualization. **Enoch Yi-No Kang:** Methodology, Formal analysis, Data curation. **Sophia H. Hu:** Writing – review & editing, Writing – original draft, Supervision, Resources, Methodology, Conceptualization. **Chao-Chun Huang:** Project administration, Data curation. **John M. O'Donnell:** Writing – original draft, Methodology, Conceptualization. **Jen-Chieh Wu:** Writing – original draft, Resources, Methodology, Data curation, Conceptualization.

Declaration of Competing Interest

On behalf of all of the authors. I declare that we have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this article.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.nepr.2026.104827.

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